Addendum: Scoring the Real Uploaded Dataset

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Why this file exists

The originally supplied raw data (data/influencers_real.csv, 5,000 profiles) has no 'is_fraud' column -- there is no ground truth for this population anywhere in the project. The technical and business reports validate the methodology on a separate, clearly-labeled SYNTHETIC reference set with genuine (mechanistically generated) fraud patterns, achieving 1.00 precision / 0.98 recall on a held-out split of that reference set. That is a real, honest validation of the METHOD. It is not, and cannot be presented as, a validation of accuracy on your real population, because no one has ever confirmed which of your real 5,000 profiles are actually fraudulent.

What was done instead

All 5,000 real profiles were scored with an unsupervised anomaly ensemble fit directly on this data (no labels required, so this is on solid ground) and reported as a percentile rank within the population, not a probability -- a percentile is honest about what an unlabeled anomaly score can support; a probability would imply a calibration that was never measured here. A second, clearly-marked 'transfer_fraud_probability_UNVALIDATED' column applies the Random Forest trained on the synthetic reference set to this real data, for comparison only. Where the two signals disagree sharply (53 profiles), a signals_disagree_flag is raised -- those are the best manual-audit candidates precisely because two independently-built signals couldn't agree on them.

Vetting Action Breakdown (real data, percentile-based)

Vetting Action	Count
APPROVE - Not a statistical outlier	4251
CAUTION - Top 15% anomaly, monitor 2-4 weeks	499
MANUAL REVIEW - Top 5% anomaly	199
REJECT - Top 1% anomaly, escalate for manual audit	51

Recommended next step

No verified fraud outcomes exist for this population. Before any automated rejection is driven by this file, manually audit a sample of ~150-300 profiles (oversampling the REJECT/MANUAL REVIEW buckets) to build a real labeled set, then re-run fraud_detection.py directly on that labeled file to get a genuinely validated precision/recall for THIS population.